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Branding is what people say about you when you are not in the room.

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-Jeff Bezos

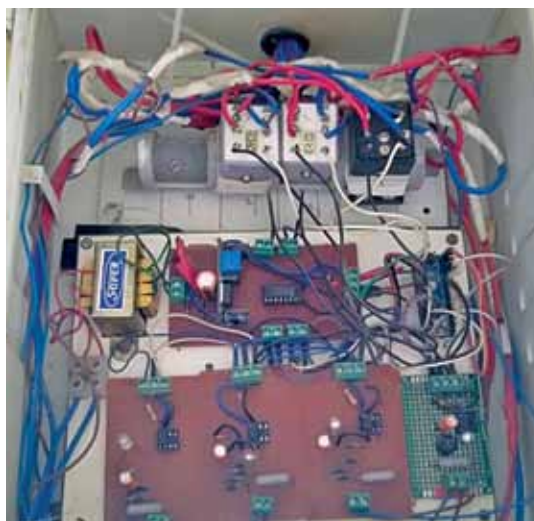


Fig. 6: The whole device setup in the building for different houses

and port name. Fig. 3 shows a code snippet with pin definitions for inputs and outputs, while Fig. 4 displays a code snippet with time lapse logic for each house.

Construction and working

Begin by uploading the source code 'rentedhouses4.ino' to the Arduino Nano using a laptop or desktop computer. Assemble the circuit on a breadboard or a general-purpose PCB, and connect the Arduino Nano board with the uploaded source code to it. Provide an unregulated supply of 15V to 18V DC across the Vcc and GND. Double-check your circuit connections to match the schematic diagram shown in Fig. 2.

For the control circuit's 230V AC power supply, use a common source, preferably the house owner's supply (if different from the four rented houses). This allows the owner to control the motor's operation. Ensure that only one relay operates at any given time, with a delay of 60, 61, 62, or 63 seconds based on house priority. After confirming this, proceed with the power circuit wiring as per the schematic diagram in Fig. 2. Both power and control circuits should be assembled and wired within a metal enclosure, placed conveniently close

to the motor or the houses (consider cable lengths), and connect the input and output motor power cables.

Testing the project is straightforward. After connecting, energise only the control circuit's 230V AC supply and test the operation of output relays by providing 230V AC supply to opto inputs one by one initially and then simultaneously to all opto inputs afterward. At any moment, only one relay should operate, with a delay of 60, 61, 62, and 63 seconds, based

on house priority. Fig. 5 shows the water valve and water motor setup for the houses, while Fig. 6 shows the complete device setup in the building for different houses.

Precautions

1. Do not forget to connect a safety earth cable for the metal enclosure (use two separate earth wires if a 3-phase supply system is involved).
2. Power fuses for short-circuit protection (glass/ceramic) should be rated for 500V AC if the supplies are from different single phases (R, Y, B) of a 3-phase system; otherwise, 250V AC is sufficient.
3. 12V DC relay contacts are rated for 6A AC, so ensure the motor's current rating does not exceed 6A.
4. Additional features, such as motor ON indication near each house's motor switch and automatic operation of pump motor discharge valves with solenoid valves, can be added, but this would involve additional cost and wiring, which is not included in this design. **EFY**



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